Cloud Production Infrastructure SOP

1. Production Deployment Workflow

All cloud production services are deployed via GitOps declarative pipelines in GitLab CI/CD. Direct manual mutations using cloud console portals are strictly prohibited.

Production releases must undergo static application security testing (SAST), automated unit tests (>85% coverage), and container image vulnerability scans prior to deployment approval.

Every production deployment requires approval from at least two senior code reviewers and an automated sanity run in staging.

2. Canary Release Strategy

New application versions are deployed using progressive traffic canary routing:

Step 1: Direct 5% of inbound user traffic to the canary pod cluster for 15 minutes.

Step 2: Monitor HTTP 5xx error rate and p99 latency metrics. If error rate remains below 0.05%, advance traffic split to 25%.

Step 3: After 30 minutes of stable metrics, route 100% of production traffic to the new version.

3. Automated Rollback Procedures

Automated rollback triggers immediately if canary error rates exceed 0.2% over a rolling 3-minute window, or if p99 latency exceeds 450 milliseconds.

The CI/CD orchestrator reverts DNS routing to the stable green deployment and drains canary connections within 60 seconds.

Post-rollback, the on-call Site Reliability Engineer must freeze production pipelines and initiate an incident post-mortem within 24 hours.

4. Disaster Recovery Targets (RTO / RPO)

Recovery Time Objective (RTO): Production core API services must restore normal operations within 60 minutes of catastrophic regional outage.

Recovery Point Objective (RPO): Database transaction logs are continuously replicated across dual regions with maximum permissible data loss window of 5 minutes.